

# AHMED AADHIL

Padukka, Colombo District, Sri Lanka

[aslamaadhil99@gmail.com](mailto:aslamaadhil99@gmail.com) | [+94 71 283 3936](tel:+94712833936) | [aadhilaslam88.github.io/Portfolio/](https://aadhilaslam88.github.io/Portfolio/)  
[linkedin.com/in/ahmed-aadhil](https://linkedin.com/in/ahmed-aadhil) | [github.com/AadhilAslam88](https://github.com/AadhilAslam88)

---

## SUMMARY

Motivated B.Sc. (Hons) Data Science student (GPA 3.09) with hands-on experience in Python, SQL, computer vision, and cloud-based systems. Built end-to-end projects including a Real-Time MLOps Drift Detection & Monitoring System, a Customer Churn Prediction Platform, a Traffic Red-Light Violation Detection System, a Cloud-Based Hostel Entry/Exit Management System, and a Face Recognition System using tools like OpenCV, NumPy, scikit-learn, XGBoost, MySQL, Docker, and Firebase/AWS. Strong foundation in statistical modeling, machine learning, data engineering, MLOps, and NLP, with proven ability to design, implement, and deploy analytical solutions. Seeking a Data Science / Machine Learning / Data Engineering / Data Analytics internship to apply predictive analytics, data visualization, and cloud skills to real-world business problems.

---

## PROJECTS

### Real-Time MLOps Drift Detection & Monitoring System

*Aug 2026 – Present*

- Building an end-to-end production model-monitoring system to detect Data Drift, Concept Drift, and Prediction Drift in machine learning models post-deployment.
- Designing architecture that bridges static model training with dynamic real-world environments to ensure continuous model reliability.
- Implementing automated alerting and response workflows to flag and address model performance decay.
- Applying statistical drift-detection methods and building a monitoring dashboard for real-time visibility into model health.

### Customer Churn Prediction Platform

*Jul 2026 – Jul 2026*

- Built an end-to-end ML pipeline that ingests CRM data, trains and evaluates churn-prediction models, and serves predictions through a REST API.
- Developed an interactive Streamlit dashboard for business users to explore churn risk and model outputs in real time.
- Containerized the full stack with Docker for reproducible training and deployment, mirroring a production ML-serving workflow.
- Applied feature engineering and model evaluation (XGBoost vs. baseline classifiers), plus API design — directly relevant to ML systems and model-deployment work.

### Traffic Red-Light-Running Violation Detection System

*Jul 2025 – Oct 2025*

<https://github.com/AadhilAslam88/Traffic-Red-Light-Running-Violation-Detection>

- Developed a computer vision system using Python and OpenCV to detect vehicles running red-light signals from video feeds automatically.
- Applied image processing techniques for vehicle detection, tracking, and violation-frame capture.
- Integrated with a MySQL database to log violation records with timestamps and evidence images.
- Demonstrated practical application of computer vision and database management in a traffic-enforcement use case.

### Retail Analytics Dashboard

*Aug 2025 – Sep 2025*

- Built a Power BI dashboard using transactional retail data to analyze sales trends, top products, and revenue by category/region.
- Wrote SQL queries for data cleaning, aggregation, and feature extraction to support visual analytics and KPIs.
- Demonstrated skills in data wrangling, BI reporting, and decision-support dashboards.

### Cloud-Based Hostel Entry-Exit Management System

*Feb 2025 – May 2025*

<https://sltc-unigo.netlify.app/>

- Designed and proposed a cloud-based hostel management system for real-time monitoring of student entry/exit activities.
- Implemented QR-code-based check-in/out using university ID cards to reduce manual logging and improve accuracy.
- Utilized cloud databases (Firebase/AWS) for secure, scalable data storage and real-time dashboard analytics for wardens/admins.

- Built a notification system to alert administrators of suspicious or unauthorized exits, enhancing security and accountability.

## Face Recognition System

May 2024 – Jun 2024

<https://github.com/AadhilAslam88/facerecognition>

- Developed a real-time face recognition application using Python and OpenCV for face detection and identification.
- Integrated with an online MySQL database to manage user profiles and recognition logs (user ID, time, recognition status).
- Demonstrated full-stack development by combining front-end logic (image capture, recognition) with back-end data storage and retrieval.

## EDUCATION

---

### B.Sc. (Hons) in Data Science

2023 – 2027

SLTC Research University (Sri Lanka Technology Campus) — Colombo, Western Province | GPA: 3.09

Key Coursework: Python Programming, Data Structures & Algorithms, Statistics, Linear Algebra, Machine Learning, SQL, Data Visualization, Data Warehousing, Natural Language Processing, Cloud Computing, Cloud Security & Privacy, Big Data.

## SKILLS

---

### Technical Skills

- **Programming Languages:** Python, SQL, R, Java, C
- **Data Science & Machine Learning:** NumPy, pandas, scikit-learn, XGBoost/LightGBM, OpenCV, classification, regression, clustering, feature engineering, model evaluation, hyperparameter tuning, statistics, probability, time-series analysis
- **MLOps & Model Deployment:** Model monitoring, data/concept/prediction drift detection, REST API model serving (Flask/FastAPI), Docker containerization, CI/CD basics, version control for ML (Git, DVC concepts), experiment tracking
- **Data Engineering:** ETL/ELT pipelines, Data Warehousing (Star/Snowflake schema), Big Data concepts (Hadoop/Spark ecosystem), data pipeline orchestration concepts (Airflow), data cleaning & validation
- **Data Analysis & BI:** Power BI, Excel, KPI design, dashboarding, business reporting, data storytelling, exploratory data analysis (EDA), A/B testing fundamentals
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Power BI
- **NLP & Computer Vision:** NLTK, text mining, tokenization, sentiment analysis, OCR (easyOCR, pytesseract), OpenCV
- **Databases:** MySQL, relational modeling, SQL (DDL/DML, joins, aggregations, window functions)
- **Cloud & Security:** AWS, Firebase, cloud computing fundamentals, cloud security & privacy, access control, data governance basics
- **Tools & Web:** Git, GitHub, Docker, Jupyter Notebook, VS Code, Streamlit, HTML, CSS

### Soft Skills

- Problem-Solving & Critical Thinking
- Teamwork & Collaboration
- Leadership & Initiative (team projects & competitions)
- Time Management & Adaptability
- Attention to Detail
- Fluent in English

## REFERENCE

---

### Dr. Chameera De Silva, PhD (UniMelb)

Head of the Department – Data Science / Lecturer

FOCIT – Faculty Industry Placement Coordinator

Member of Ethics Review Committee

Member of Technical Review Committee

Sri Lanka Technology Campus (SLTC), Padukka, Sri Lanka

Mobile: +94 70 159 9764

Email: [chameera.d@sltc.ac.lk](mailto:chameera.d@sltc.ac.lk) | [hod.ds@sltc.ac.lk](mailto:hod.ds@sltc.ac.lk)

Web: [www.sltc.ac.lk](http://www.sltc.ac.lk)

P. Email: [chameeradesilva@acm.org](mailto:chameeradesilva@acm.org) | [chameeradesilva@sigchi.org](mailto:chameeradesilva@sigchi.org)

Personal Web: [www.chameeradesilva.com](http://www.chameeradesilva.com)